



Fall semester 2022-2023: Math 102 Calculus II
First Exam: Suggested Solutions

Q1: [10 pts.] Find $\lim_{x \rightarrow \infty} (x + e^x)^{\frac{2}{x}}$, if it exists.

The limit is indeterminate of the form ∞^0 . So let us assume that $y = (x + e^x)^{\frac{2}{x}}$. Then

$$\ln y = \ln (x + e^x)^{\frac{2}{x}} = 2 \frac{\ln (x + e^x)}{x}.$$

Thus

$$\lim_{x \rightarrow \infty} \ln y = 2 \lim_{x \rightarrow \infty} \frac{\ln (x + e^x)}{x},$$

which is an indeterminate form of type ∞/∞ , so by using l'Hospital's rule three times we obtain

$$\lim_{x \rightarrow \infty} \ln y = 2 \lim_{x \rightarrow \infty} \frac{1 + e^x}{x + e^x} = 2 \lim_{x \rightarrow \infty} \frac{e^x}{1 + e^x} = 2 \lim_{x \rightarrow \infty} \frac{e^x}{e^x} = 2.$$

This implies that $\lim_{x \rightarrow \infty} (x + e^x)^{\frac{2}{x}} = e^2$.

Q2: [15 pts.] a) Evaluate $\int \frac{\cos x}{\csc 3x} dx$.

Using the trigonometric identity $\sin A \cos B = \frac{1}{2} [\sin(A - B) + \sin(A + B)]$. The integral can be written as follows:

$$\begin{aligned} \int \frac{\cos x}{\csc 3x} dx &= \int \sin 3x \cos x dx \\ &= \frac{1}{2} \int (\sin 2x + \sin 4x) dx = -\frac{\cos 2x}{4} - \frac{\cos 4x}{8} + C. \end{aligned}$$

b) Evaluate $\int \frac{1}{\sqrt{x-1}(x+3)} dx$.

Using the rationalization substitution $u = \sqrt{x-1}$, thus $du = \frac{1}{2\sqrt{x-1}} dx = \frac{1}{2u} dx$, then we have

$$\begin{aligned} \int \frac{1}{\sqrt{x-1}(x+3)} dx &= \int \frac{2u}{u(u^2+4)} du = 2 \int \frac{1}{u^2+4} du \\ &= \frac{1}{2} \int \frac{1}{(\frac{u}{2})^2+1} du = \arctan\left(\frac{u}{2}\right) + C = \arctan\left(\frac{\sqrt{x-1}}{2}\right) + C. \end{aligned}$$

Q3: [10 pts.] Evaluate $\int \frac{4x^2 + 1}{x^3 + 2x^2} dx$.

Using the partial fraction decomposition gives

$$\frac{4x^2 + 1}{x^3 + 2x^2} = \frac{4x^2 + 1}{x^2(x + 2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x + 2},$$

$$4x^2 + 1 = Ax(x + 2) + B(x + 2) + Cx^2,$$

therefore, we obtain $A = -\frac{1}{4}$, $B = \frac{1}{2}$ and $C = \frac{17}{4}$ which implies

$$\begin{aligned} \int \frac{4x^2 + 1}{x^3 + 2x^2} dx &= -\int \frac{1}{4x} dx + \int \frac{1}{2x^2} dx + \int \frac{17}{4(x + 2)} dx \\ &= -\frac{1}{2x} - \frac{1}{4} \ln |x| + \frac{17}{4} \ln |x + 2| + C. \end{aligned}$$

Q4:- [10 pts.] Evaluate $\int \tan^2 x \sec^4 x dx$.

Since $\frac{d}{dx} \tan x = \sec^2 x$, we rewrite the integral as

$$\int \tan^2 x \sec^4 x dx = \int \tan^2 x \sec^2 x \sec^2 x dx = \int \tan^2 x (1 + \tan^2 x) \sec^2 x dx$$

Now, we let $u = \tan x$, so that $du = \sec^2 x dx$ and

$$\begin{aligned} \int \tan^2 x \sec^4 x dx &= \int \underbrace{\tan^2 x (1 + \tan^2 x)}_{u^2(1+u^2)} \underbrace{\sec^2 x dx}_{du} \\ &= \int u^2(1 + u^2) du = \int (u^2 + u^4) du \\ &= \frac{1}{3} u^3 + \frac{1}{5} u^5 + C = \frac{1}{3} \tan^3 x + \frac{1}{5} \tan^5 x + C. \end{aligned}$$

Q5: [15 pts.] Evaluate $\int \sqrt{1 - \frac{4}{x^2}} dx$, for $x \geq 2$.

We note that $\sqrt{1 - \frac{4}{x^2}} = \frac{\sqrt{x^2 - 4}}{x}$. So, we can take $x = 2 \sec \theta$, for $\theta \in \left[0, \frac{\pi}{2}\right)$, and thus $dx = 2 \sec \theta \tan \theta d\theta$.

$$\begin{aligned} \int \sqrt{1 - \frac{4}{x^2}} dx &= \int \frac{\sqrt{x^2 - 4}}{x} dx \\ &= \int \frac{2 \tan \theta}{2 \sec \theta} \cdot 2 \sec \theta \tan \theta d\theta \\ &= 2 \int \tan^2 \theta d\theta \\ &= 2 \int (\sec^2 \theta - 1) d\theta \\ &= 2[\tan \theta - \theta] + C \\ &= \sqrt{x^2 - 4} - 2 \operatorname{arcsec} \left(\frac{x}{2}\right) + C \end{aligned}$$

Q6:- [15 pts.] Evaluate $\int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{\sin x}{2 \sin x - \sin 2x} dx$.

Since $\sin 2x = 2 \sin x \cos x$, the integral can be written as follows:

$$\int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{\sin x}{2 \sin x - \sin 2x} dx = \frac{1}{2} \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{1}{1 - \cos x} dx.$$

Using the Weierstrass substitution $t = \tan(x/2)$, and if $x : \pi/2 \rightarrow 2\pi/3$, then $t : 1 \rightarrow \sqrt{3}$. Thus we have

$$\begin{aligned} \int_{\frac{\pi}{2}}^{\frac{2\pi}{3}} \frac{\sin x}{2 \sin x - \sin 2x} dx &= \frac{1}{2} \int_1^{\sqrt{3}} \frac{1}{1 - \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt \\ &= \frac{1}{2} \int_1^{\sqrt{3}} \frac{1}{t^2} dt = \frac{1}{2} \left[-\frac{1}{t} \right]_1^{\sqrt{3}} = \frac{1}{2} \left[1 - \frac{1}{\sqrt{3}} \right]. \end{aligned}$$

Q7:- [15 pts.] Evaluate $\int x^3 \sin x^2 dx$.

Let $z = x^2$ then $dz = 2x dx$, so the integral can be written as follows:

$$\int x^3 \sin x^2 dx = \frac{1}{2} \int z \sin z dz.$$

Then, integrating by parts with $u = z$ and $dv = \sin z dz$, leads to:

$$\begin{aligned} \int x^3 \sin x^2 dx &= \frac{1}{2} \left[-z \cos z + \int \cos z dz \right] \\ &= \frac{1}{2} [-z \cos z + \sin z] + C = \frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + C. \end{aligned}$$

Q8: [10 pts.] Determine whether the integral $\int_2^3 \frac{dx}{\sqrt{3-x}}$ is convergent or divergent.

The given integral is improper because $f(x) = 1/\sqrt{3-x}$, has the vertical asymptote $x = 3$. Since the infinite discontinuity occurs at the right endpoint of $[2, 3]$, then if $u = 3 - x$ and $du = -dx$, we have

$$\int \frac{dx}{\sqrt{3-x}} = \int -u^{-1/2} du = -2\sqrt{u} + C = -2\sqrt{3-x} + C,$$

and

$$\int_2^3 \frac{dx}{\sqrt{3-x}} = \lim_{t \rightarrow 3^-} \int_2^t \frac{dx}{\sqrt{3-x}} = -2 \lim_{t \rightarrow 3^-} \sqrt{3-x} \Big|_2^t = -2 \lim_{t \rightarrow 3^-} [\sqrt{3-t} - 1] = 2,$$

it means that the improper integral is convergent.